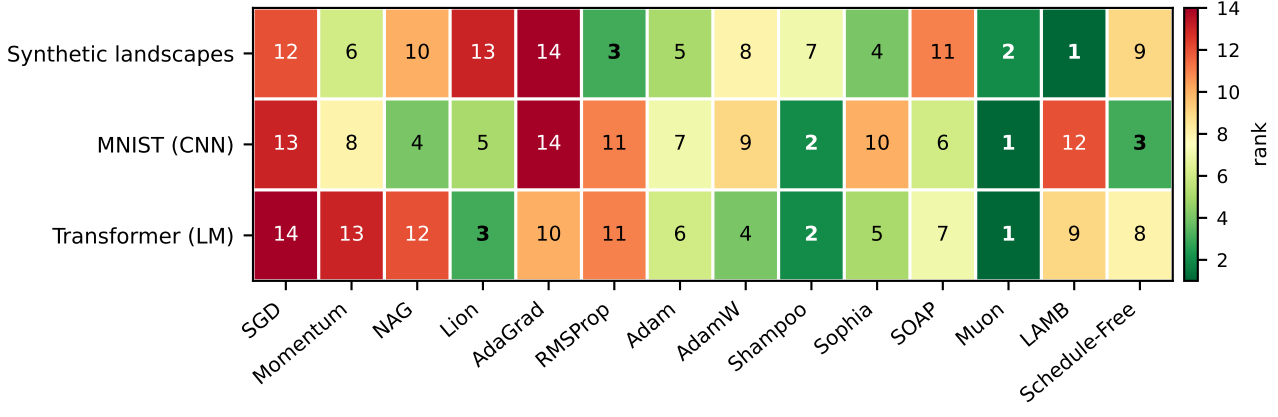
Synthetic landscapes
MNIST (CNN)
Transformer (LM)
12 6 10 13 14 3 5 8 7 4 11 2 1 9
13 8 4 5 14 11 7 9 2 10 6 1 12 3
14 13 12 3 10 11 6 4 2 5 7 1 9 8
SGD
Momentum
NAG
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
Schedule-Free
rank
14
12
10
8
6
4
2